

Institute of Actuaries of India

Subject SP7 – General Insurance Reserving and Capital Modeling

May 2023 Examination

INDICATIVE SOLUTION

Introduction

The indicative solution has been written by the Examiners with the aim of helping candidates. The solutions given are only indicative. It is realized that there could be other points as valid answers and examiner have given credit for any alternative approach or interpretation which they consider to be reasonable.

Solution 1:

Assumptions:

1. Company is only 5 years old or writing Motor TP business from 5 years and data is not available before 5 years
2. AY 2018-19 is not fully developed
3. Difference in ultimate of and incurred of AY 2018-19 is tail factor
4. For calculation of discounted reserve, it is assumed that, runoff happens in 6 years
5. Incurred triangle will be more appropriate in calculating of development of IBN(E)R than paid triangle as future development in Incurred triangle is just because of IBN(E)R.

[0.5 Marks for each point, Max 2 Marks]

Accident year / Dev year	Ultimate (1)	ITD (2)	CDF =(2)/(1)	DF
2018-2019	6270000	5700000	1.10	1.10
2019-2020	6279000	4830000	1.30	1.18
2020-2021	6915000	4610000	1.50	1.15
2021-2022	8122500	3610000	2.25	1.50
2022-2023	8880000	1110000	8.00	3.56

[5 marks for calculations of CDFs and DFs]

Completing the future development of incurred triangle with above calculated development factor

Accident year / Dev year	1	2	3	4	5	Ultimate amount
2018-2019	820000	2760000	3980000	4860000	5700000	6270000
2019-2020	780000	2420000	3760000	4830000	5708182	6279000
2020-2021	830000	2890000	4610000	5319231	6286364	6915000
2021-2022	1010000	3610000	5415000	6248077	7384091	8122500
2022-2023	1110000	3946667	5920000	6830769	8072727	8880000

IBNR development in Year 2023-24

$$=6270000+5708182+531923+5415000+3946667-5700000-4830000-4610000-3610000-1110000=6799079$$

Similarly, calculation IBNR development for future years will provide

Year	2023-24	2024-25	2025-26	2026-27	2027-28
IBNR development	67,99,079	43,44,361	26,75,420	19,80,367	8,07,273

IBNR development is considered at the middle of the year for calculation of discounted reserve

Tenure	Year	Undiscounted IBNR	Discounted IBNR
0.5	2020-21	6799079	6635222
1.5	2021-22	4344361	4037774
2.5	2022-23	2675420	2368202
3.5	2023-24	1980367	1669487
4.5	2024-25	807273	648139
Total		16606500	15358824

[8 marks for the computation of discounted and undiscounted IBNR]

[15 Marks]

Solution 2:

1. If claims are grouped by a reporting year cohort...
the chain ladder projection will not allow for pure IBNR, that is, for claims that are notified after the end of the reporting year.
A projection based on reporting year cohort data will only project claims that have been reported to the insurer.
This will project IBNER claims and difference between Total reserve calculated from other method and this amount will provide pure IBNR claim amount.
[1.5]
2. Accident year claim count triangles will allow for the pure IBNR. . .
Projection of same will provide delayed reporting claims at a valuation date... Multiplying delayed claim number by Ultimate ACPC would provide pure IBNR....
Once pure IBNR is evaluated, difference between that and projected ultimate by other method would be IBNER.
[1.5]

Challenges:

1. Deciding proper future Inflation adjustment
2. Impact of large and attritional claims
3. Availability of data at details needed for this exercise

[2]

[5 Marks]**Solution 3:****Data Risk:**

This is a new line for the company and enough data may not be available to price risk i.e. non availability of all rating factors

Data may be provided with Tender but just current year data which is still running

There may be error in data provided to the company

Data may be available but at summary level not at granular level to price risk appropriately

Mitigation strategy:

Adding suitable contingency loading or margin in pricing

Company may avoid quoting where complete data is not available

Company may use data of other state with proper adjustment for difference in risk but it may further increase risk

May be available with some intermediaries

Underwriting Risk:

As it is Tender so those who quote lowest premium will get business which may lead to taking optimistic view in pricing and leading losses. Further, Company do not have any prior underwriting experience in these types of schemes which increases risk.

Mitigation strategy:

1. Hiring Experienced Underwriters from the market with experience in these types of schemes
2. Writing only tender with loss corridor to limit losses
3. Appropriate reinsurance arrangements
4. Going with actuarial pricing

Administration risk

Risk of error or failure associated with the administrative aspects of firm's operations...

Need to hire experienced doctors, claims teams and operational staff in large numbers

Such large number individual may not be available in short time

Company need significant office space for such a big team which may be costly

Mitigation strategy:

Company may utilize services of third party outsourcing agency but which increases another type of risk.

Can initially quote for states where it has already crop Insurance which increases synergy in administration

Reputational Risk:

Delay in settlement of claims, will give bad advertisements

Deficiency in services of outsourcing agency may lead to penalty to Insurance company

As these schemes has political significance, company may be required to less stringent in claim settlement

Mitigation strategy:

Regular claim monitoring needs to be done

Liquidity Risk

Company only gets 40% of premium for first 6 months which may not be sufficient to pay claims for 6 months...

Further, there can be delay in obtaining second and third instalments...

... therefore need to have availability of additional liquid fund

Mitigation risk

Company needs to adequately maintain cash or short term assets

Solvency Risk:

If company do not get enough business, it will increase management of expenses with so many people on board

Company needs to refund premium retaining just small amount of administrative expenses where loss ratio is favourable

... and take complete losses on net for loss making states

...therefore, it may be high risk and less reward strategy

Company may be asked to extend policy for a few months after 1 year of loss-making policy which may further increase losses

Due to bulk premium size, one wrong pricing can endanger solvency

Company don't have enough margin between current solvency and minimum required solvency. therefore, any loss-making tender may lead to action by the regulator

Mitigation strategy:

Company may utilize services of third-party outsourcing agency which can be only used if any tender is awarded. There will cost in proportion of business

Quoting with suitable margin in pricing

Technological risk-

Developing new System to write Health Business...

.. Integrating different systems and Own system to SHA

risk increases as a result of new business lines being written so risk of adequate information not being captured...

Mitigation strategy:

Thorough study of Competitors Technology

Compliance risk... ..

risk of non-adherence to legislative guidelines of SHA as company is first time writing business and internal firm requirements. ...

risk increases as a result of new business strategies as complete processes / requirements may not be known

There is significant penalty for non-adherence to any of clause in these type of tender

Mitigation strategy:

On boarding Compliance officer who has experience of working with company writing Government Health mass scheme

Event risk.

...significant event such as pandemic that occurs and has significant impact on firm's operations. ... for example, a pandemic where due to significant flow of claims cannot be processed and resulting in penalty and reputational damage

Fraud risk...

...risk associated with the fraudulent practices of hospital and insured raising fraudulent claims ...

Risk associated with fraudulent practices of employees increases due to new team and additional personnel

...risk associated with senior management not effectively performing roles.

Mitigation strategy:

Fraud Analytics to detect fraud and taking harsh action

Purchasing Fraud monitoring software's

Strategic risk...

...risk arising from the inability to implement appropriate business plans and strategies, make decisions, such as

allocate resources or adapt to changes in the business environment ...both Governance and Strategic are an increased risk due an entirely new Health team who have embarked upon strategic change which has not yet been evaluated

[20 Marks]

Solution 4:

i) Assumptions

- All policies are earned over a one year period
- The terms of the quota share have remained constant over the period 1.1.2019 to 31.12.2022
- As the company began trading on 1.1.2019 all premium earned in 2019 was written in 2019
- All business written in a year has the same loss ratio

[3]

premium is earned 2/5 in the year it is written and 3/5 in the next year

For 2019:

Net claims on Accident year= $10000 \times 1.5 = 15000$ and Net claims on underwriting year= $25000 \times 1.5 = 37500$. Net loss ratio on u/w basis= 150%

For 2020:

Net claims on Accident year= $35000 \times 130\% = 45500$ as $37500 - 15000 = 22500$ of these were on business written in 2019, then 23000 claims occurring in 2020 on 20000 earned premium, i.e. claim ratio of 115%.

Net loss ratio on u/w basis= 115% and Net claims on underwriting year= $50000 \times 1.15 = 57500$

For 2021

Similarly, Net claims on Accident year= 70000

57500-23000=34500 in respect of business written in 2020 thus 35500 for business written in 2021 on 40000 premium, hence Net loss ratio on u/w basis=88.75% and Net claims on underwriting year= 88750

For 2022

Similarly, Net claims on Accident year= 105000

887500-35500=53250 in respect of business written in 2021 thus 51750 for business written in 2022 on 80000 premium, hence Net loss ratio on u/w basis=64.69 and Net claims on underwriting year= 129375

	2019	2020	2021	2022
Net claims on AY	15,000	45,500	70,000	105,000
Net claims on UW	37,500	57,500	88,750	129,375
Net Written LR	150%	115%	88.75%	64.69%

[4, for all correct numbers of each year]

As QS reinsurance is assumed to remain constant the GWLR = NWLR

[1]

[8]

ii)

- Rapidly decreasing loss ratio and year on year growth are unlikely to happen at the same time
 - Household insurance is a commodity market and so usually sells on price
 - So to build a book quickly you have to be cheaper than the competition
- Override or ceding commissions may mean that net ULR does not equal gross ULR
- Unrealistic that change premiums only once a year
- Percentage reinsured likely to change over the years
- Household profits tend to fluctuate with the occurrence/non-occurrence of weather events
- Specific assumptions in the question not being realised in practice
- So smooth increase in profits is unlikely

[5]

iii)

- Fast growth only available by writing business at premiums less than the market (unprofitably?) or having a unique customer proposition.
- Reserve for claims / premiums would increase especially if rate are unprofitable
- New business strain could be considerable
- Any imposed Solvency ratio would mean capital considerations could restrict the future rate of growth. If near the Solvency ratio brokers may stop placing business, even policyholder may hear that the company is not safe
- Unless there is enough capital to comfortably support growth then plans will need to be scaled back
- May change investment strategy
- May change reinsurance strategy
- May increase staff accommodation, IT systems etc

[5]

iv)

- a) EML is the expected or estimated maximum loss The largest loss expected to arise from a single event in respect of an insured property which may well be less than its market or replacement value. Used as an exposure measure in rating certain classes, including reinsurance

[2]

b) $(1+N) * R$

[1]

- c) 10,000,000 EML is split as follows A:B:C 5:2:3 as the minimum is always ceded
 Claim is INR 9,000,000, A will pay 5,000,000, B will pay $\frac{2}{5}$ of $(9000000-5000000)= 1600000$,
 C will pay 2400000

[4]

[25 Marks]

Solution 5:

i)

- The frequency and average cost can be related to
 - the economy (Housing market)
 - several other events, e.g. the weather (unusually dry spells), closeness of trees, pipes and soil conditions.
- Long notification and settlement delays
- Low frequency and high severity
- Can have high levels of reopened claims.
- Claim size distribution can be highly skewed.
- There can be significant accumulations of claim occurrences and notifications during particular periods.
- Claim costs subject to inflation.
- Date of occurrence is usually uncertain
- This may lead to disputes about coverage especially when there have been changes in the insurance company providing the cover
- Need for individual loss assessment by a claims adjuster
- Active claim management can significantly reduce the cost of a claim.

[5]

ii)

- Industry data on claim notifications and claim costs.
- Externally available models
- Own claims experience of subsidence
- Judicial findings and market practice
- Economic data on Housing Market
- Building cost Index as a proxy for claim cost inflation.
- Climate and weather data e.g. Rainfall data, Soil moisture and composition data etc.
- Applicable Tax rates
- Expense loadings (including brokerage, commission and management expenses).
- Required Profit loadings
- Investment yields
- Cost of capital
- Contingencies
- Retrocession costs (if applicable)
- Competitors prices (if available)

[5]

[10 Marks]

Solution 6:

- i) It is caused by over-supply of insurance capacity in certain products. Premium rates start to 'soften' during this phase as insurers compete for market share. Eventually premium rates fall to the extent that insurance is generally loss-making. This position is commonly known as a soft market.

[1]

The soft market is a term describing that part of the insurance cycle when business is least profitable. i.e. inadequate premiums being paid for the amount of risk assumed. Its longevity and depth result from the true effects of claims inflation, broadening wordings cover, and increases in exposure being under-estimated at the time of writing business. Persists because premium increases lag behind claims notifications.

Four factors that encourage the persistence of the soft market:

- the ease with which new entrants can join insurance markets
- the delay between writing business and knowing how profitable it is
- simplistic regulatory capital requirements that encourage insurers to write more business when premium rates are falling and less business when they are rising
- economies of scale, which encourage marginal costing.

[4]

[5]

ii) Risks of underestimating softness in the market:

- Write a larger volume of unprofitable business than planned leading to an overstatement of profits and Understatement of reserves.
 - managers not aware of true financial condition
 - leading to bad business and management decisions, eg- reinsurance purchasing, investment strategy.
- Under-reserving could lead
 - to a shortfall of funds and an inability to meet liabilities as they become payable
 - will speed up the payment of dividends and tax.
 - filter through to the premium rating exercise, resulting in incorrect rates.
 - An incorrect assessment of reserves may lead to inappropriate management decisions.
- Risk of writing larger volumes of business than planned and not being able to service it.
- There is a risk that the true profitability of business written is not known for some time
 - the company writes business at less profitable rates than it expects
 - Leading to a significant weakening of the financial condition of the company.
- When the extent of this position is realized reserves must be increased (sometimes dramatically)
 - leading to a reduced solvency position
 - requiring additional capital support and/or other remedial action.
- This reduced solvency may directly impact the quantity of profitable new business that the company can write and in the extreme may lead to the insolvency of the company.

[10]

[15 Marks]

Solution 7: The numbers of factors which a company needs to consider while formulating its investment strategy:

1. Liabilities of the companies

- Category of Inflation
 - Motor OD- Most of the claims will be for repair. Significant proportion of claim is labour cost which will be related general wage inflation. Total Loss and Theft claims will be fixed benefit in Nature
 - Personal Accident will be fixed benefit in Nature
 - Motor TP – Loss of Income component will be related to income related costs, personal injury claims will be related to Medical Expense inflation. There will be interest award by court and superimposed inflation.
- All Claims will be in Indian currency

- Due to significant delay in reporting and settlement of Motor TP claims, it will have significant proportion of medium term and long-term liability
- Significant uncertainty in Amount and timing of Motor TP claims
- Estimated future liabilities from business planned
- New legislations which can increase Motor TP claims retrospectively

2. Assets

- Company needs to invest significantly in long duration assets to earn higher return to meet Motor TP Interest award
- Need to invest in inflation linked investment
- Long term liabilities so can invest higher proportion of investment in equity in comparison to companies writing all LOBs
- Economic outlook
- Expected volatility within the various asset classes

3. External influences

- Tax treatment of different investments and the tax position of the company
- statutory, legal, ethical or voluntary restrictions on how the insurer may invest
- statutory valuation requirements
- solvency requirements – Margin between current solvency and regulatory minimum requirements
- rating agency constraints on capital required to maintain the insurer's desired rating
- regulatory guidelines on Investment

4. insurer-specific considerations

Risk appetite – each general insurer will differ in the amount of risk it is willing / able to accept, perhaps due to other considerations within the company group, wider management objectives or board appetite.

Company-specific investment objectives

[10 Marks]
